Dinesh Patra

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Research Statement

My broad research interests currently lie in multi-agent decision-making and control in constrained and uncertain environments, spanning cooperative, competitive, adversarial, and game-theoretic interactions. Drawing on tools from **geometric and Lyapunov-based analysis**, **model predictive control**, **safety-critical control**, and **game-theoretic control**, my doctoral research has addressed **multi-agent pursuit–evasion problems**, with a focus on designing strategies that guarantee **encirclement and finite-time capture** in constrained environments.

Education

Ph.D. in Control Systems (ongoing) Aug 2021 – Jul 2027 (Tentative)
Indian Institute of Technology Kharagpur, Kharagpur, India GPA: 8.6/10

- Tentative Thesis Title: *Encirclement and Capture Guarantees in Constrained Environment against Unknown Evader Strategies*
- Tentative Thesis Submission: February 2027
- Advisor: Prof. Ashish Ranjan Hota, Associate Professor, Department of Electrical Engineering, IIT Kharagpur
- Coursework: Estimation of Signals and Systems, Graph Theory, Nonlinear Control, Convex Optimization.

M.Tech. in Control & Industrial Automation Jul 2019 – May 2021
National Institute of Technology Silchar, Silchar, India GPA: 9.48/10

- Thesis: *A New Observer-Based Chattering-Free-Integral-Non-Singular-Terminal Sliding Mode Controller for a System with Disturbances*
- Advisor: Prof. Binoy Krishna Roy, Professor, NIT Silchar, Silchar, Assam
- Coursework: Linear Control Theory, Optimal Control, Estimation.

B.Tech. in Electrical Engineering Jul 2013 – May 2017
Parala Maharaja Engineering College, Brahmapur, India GPA: 8.51/10

Research Experience

Doctoral Research (ongoing) Aug 2021 – Present
Department of Electrical Engineering, IIT Kharagpur Kharagpur, India

- Encirclement guaranteed capture in multi-agent pursuit–evasion using geometric approaches, model predictive control, and control barrier functions; turn-rate-constrained capture; target defense; robust receding-horizon games.

Visiting Graduate Student May 2025 – Jul 2025
Michigan State University, Department of Electrical and Computer Engineering East Lansing, MI, USA

- Research collaboration with Prof. Shaunak D. Bopardikar, supported by the IEEE Control Systems Society Graduate Collaboration Fellowship.

Sponsored Project Feb 2022 – Jul 2023
Integrated Test Range (ITR), Defence Research and Development Organisation (DRDO)

- Worked on *Trajectory Prediction and Safety Assessment of a Tactical Aerial Vehicle*. PI: Prof. Ashish R. Hota; Co-PI: Prof. Siddhartha Mukhopadhyay.

Publications

JOURNAL ARTICLES

[J1] **Dinesh Patra**, Prajakta Surve, Ashish R. Hota and Shaunak D. Bopardikar, "Encirclement Guaranteed Finite-Time Capture against Unknown Evader Strategies," *IEEE Control Systems Letters*, 2026. arXiv:2603.15278

SUBMITTED

[S1] **Dinesh Patra**, Bhargav Jha and Ashish R. Hota, "Finite-Time Capture by a Dubins Pursuer Against Unknown Evader Strategies," 2026. arXiv:2610.25007

WORKING PAPERS

[W1] **Dinesh Patra**, Tanish Jain and Ashish R. Hota, "Robust Receding Horizon Games with Additive Uncertainty," 2026. arXiv:2607.04213

CONFERENCE PAPERS

[C1] **Dinesh Patra** and Ashish R. Hota, "3D Cooperative Pursuit with Guaranteed Enclosure via Robust Model Predictive Control," *2025 American Control Conference (ACC)*, Denver, CO, USA, pp. 2–7, 2025.

[C2] **Dinesh Patra**, Simran Kumari and Ashish R. Hota, "Model Predictive Control based Target Defense with Attacker Trajectory Prediction," *2024 European Control Conference (ECC)*, Stockholm, Sweden, pp. 597–602, 2024.

[C3] **Dinesh Patra**, Pulak Halder, Mukesh Roy, Shrabani Ghosh, Ashish R. Hota and Siddhartha Mukhopadhyay, "Trajectory Prediction under Sensor Bias Faults in a Tactical Aerial Vehicle," *2023 International Conference on Range Technology (ICORT)*, pp. 1–6, 2023. **(Best Paper Award)**

[C4] **Dinesh Patra** and Binoy Krishna Roy, "Robustness Test in Conventional and Integral Sliding Mode Control," *2021 International Conference on Communication, Control and Information Sciences (ICCISc)*, pp. 1–5, 2021.

Awards and Achievements

- **IEEE Control Systems Society Graduate Collaboration Fellowship**, awarded to at most ten selected graduate students worldwide each year; funded a research visit to Michigan State University. 2024
- **Best Poster Award**, Research and Innovation Day, Department of Electrical Engineering, IIT Kharagpur. 2025
- **Best Paper Award**, International Conference on Range Technology (ICORT) 2023, organized by the Integrated Test Range (ITR), DRDO. 2023
- **MHRD Fellowship**, Ministry of Human Resource Development, Government of India, awarded on the basis of Graduate Aptitude Test in Engineering (GATE) 2019 score for postgraduate studies. 2019
- **Travel Grants**
  - Reinforcement Learning Workshop, IISc Bangalore. Jan 2026
  - 2025 American Control Conference (ACC), Denver, Colorado, USA. Jul 2025
  - Learning and Control Colloquium, IIT Bombay. Sep 2024

Professional Service

Reviewer

- *Journals:* IEEE Transactions on Automation Science and Engineering; IEEE Transactions on Cybernetics; International Journal of Control; IEEE Robotics and Automation Letters; IEEE Control Systems Letters; IEEE Open Journal of Control Systems.
- *Conferences:* American Control Conference (ACC); IEEE Conference on Decision and Control (CDC); European Control Conference (ECC); Indian Control Conference (ICC); IFAC Conference on Advances in Control and Optimization of Dynamical Systems (ACODS); IEEE India Council International Conference (INDICON).

Teaching Experience

Teaching Assistant Aug 2021 – Present
Indian Institute of Technology Kharagpur Kharagpur, India

- Convex Optimization in Control and Signal Processing; Control and Electronic System Design Lab; PG Control Design Lab; UG Control Lab; Electrical Technology.

NPTEL Teaching Assistant Autumn 2025, Autumn 2026
National Programme on Technology Enhanced Learning (NPTEL), IIT Kharagpur

- *Convex Optimization in Control* (Instructor: Prof. Ashish R. Hota).

Positions of Responsibility

- **Core Committee Member**, Research and Innovation Day 2025, organized by the research scholars of the Department of Electrical Engineering, IIT Kharagpur.
- **Student Volunteer**, Indian Control Conference (ICC) 2027, IIT Kharagpur, India (to be held in January 2027).
- **Student Volunteer**, IEEE CSS Day 2022 Seminar Series, organized by the IEEE Control Systems Society Kharagpur Chapter, India.
- **Student Volunteer**, Fractional Order Systems and Their Applications (FOSTA) 2019, NIT Silchar, Assam, India.

Skills

Programming & Software MATLAB, Simulink, C++, YALMIP, IPOPT, MPT Toolbox, LaTeX
Language Proficiency English (bilingual), Hindi (professional working), Odia (native)